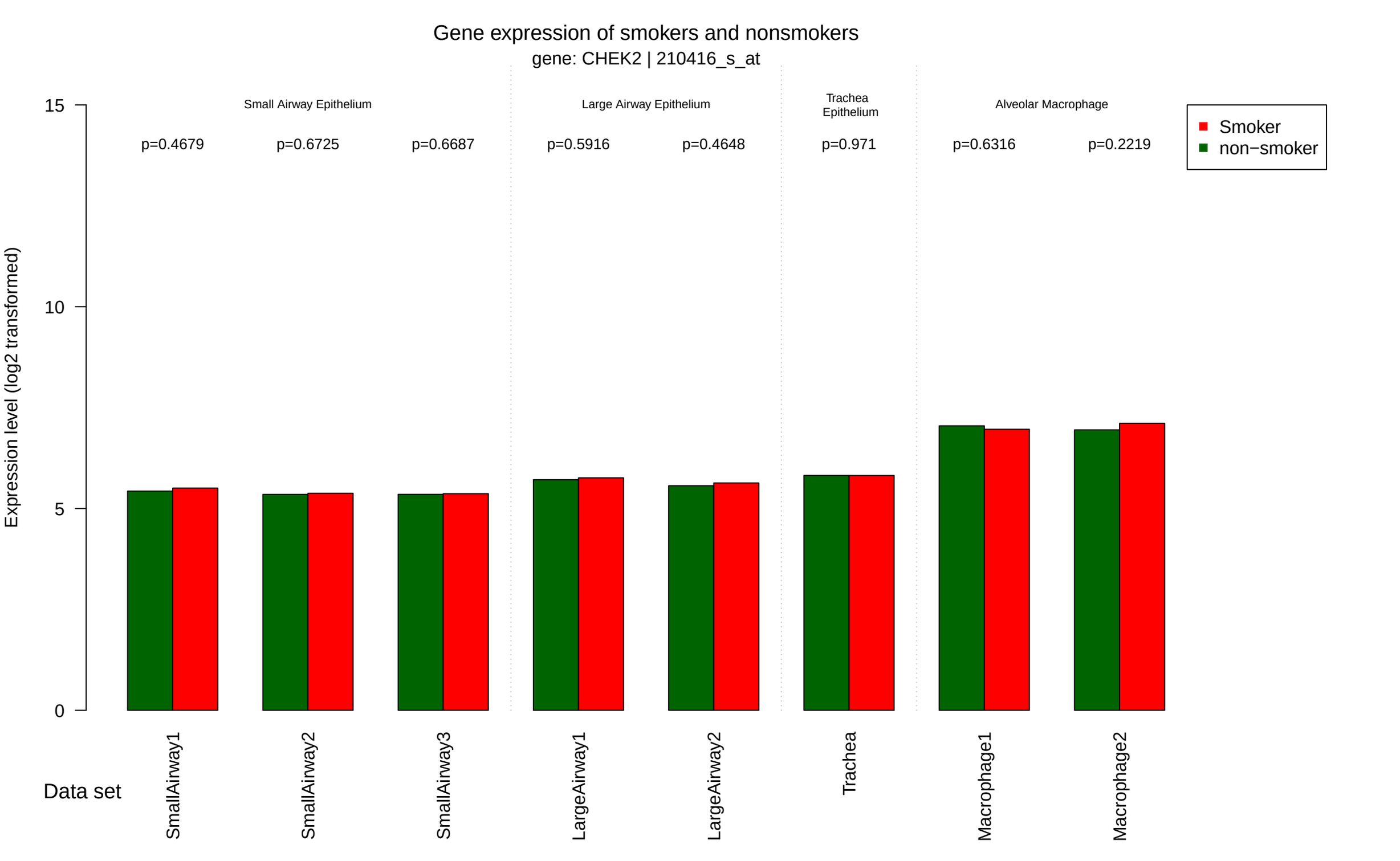
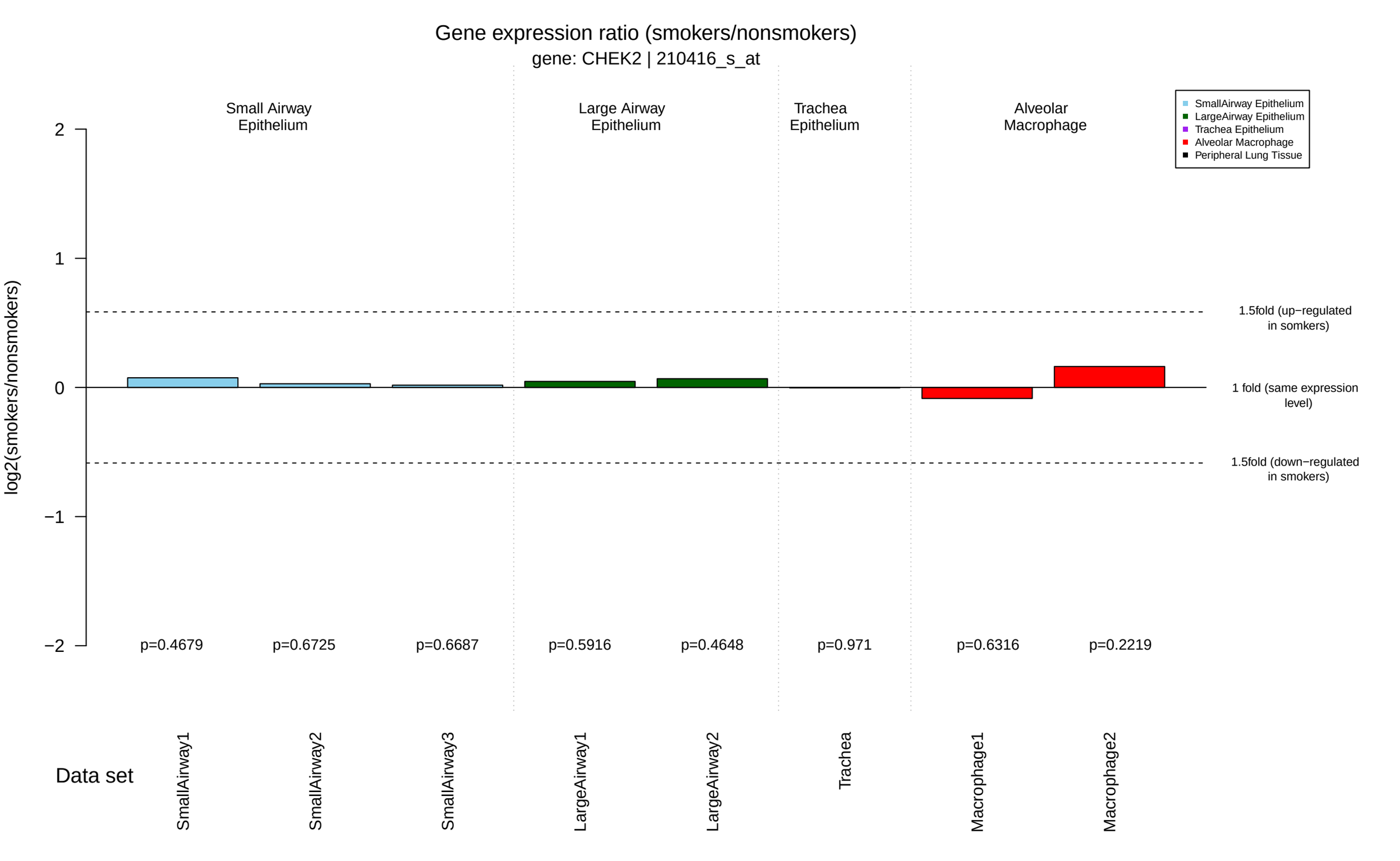
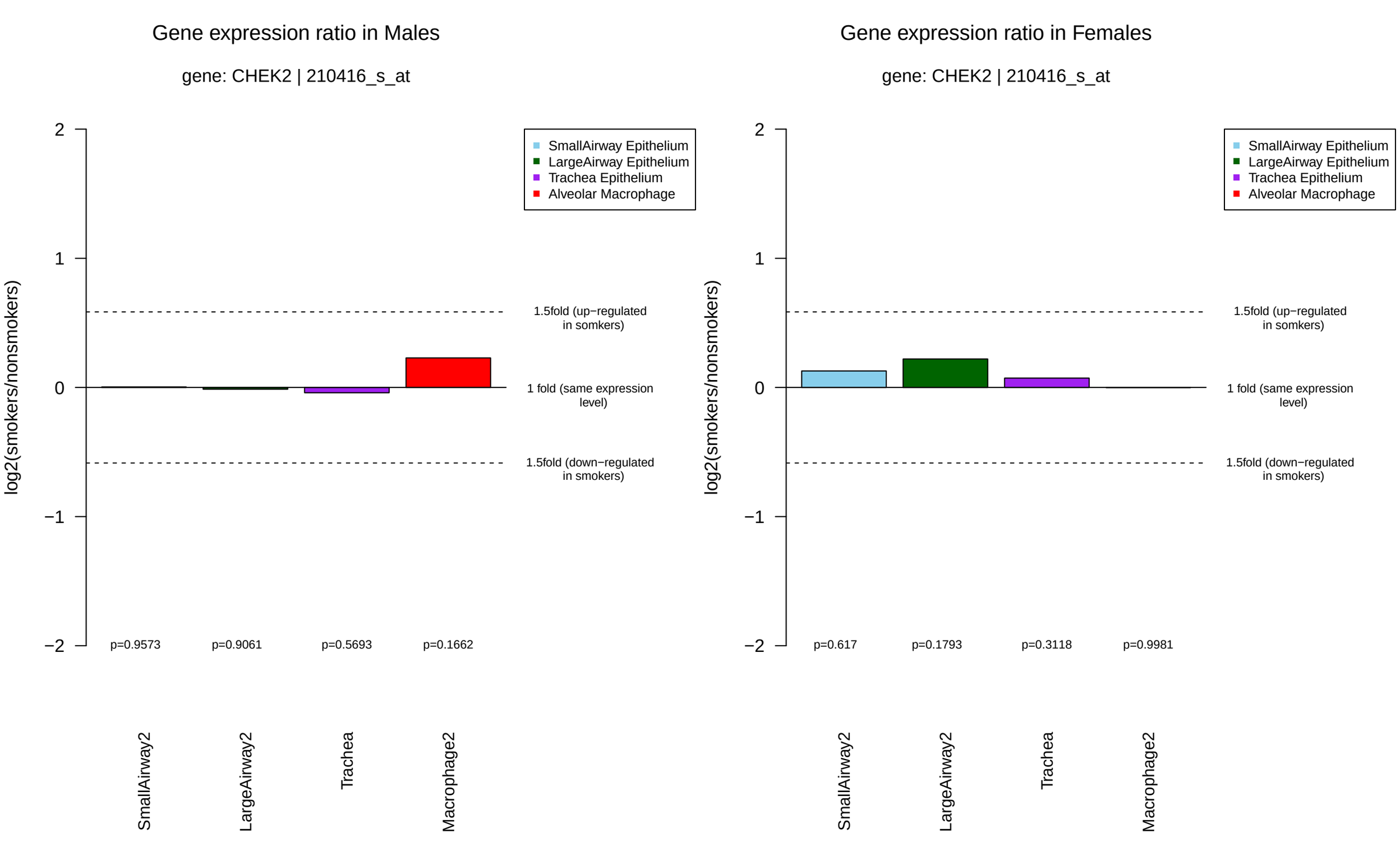
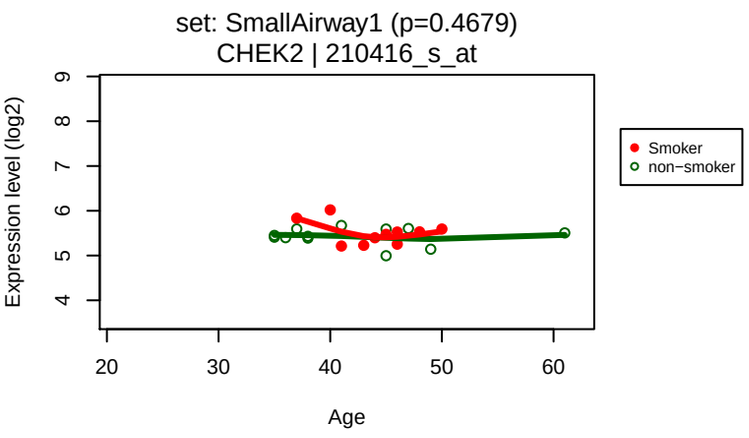
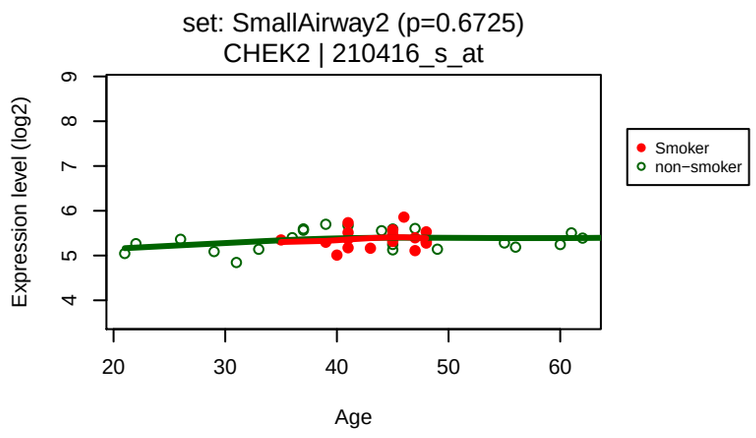
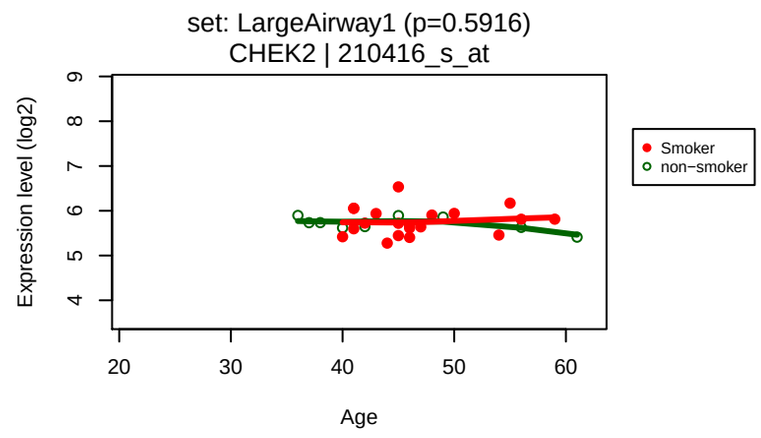

Gene expression of smokers and nonsmokers
gene: CHEK2 | 210416_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4679
p=0.6725
p=0.6687
p=0.5916
p=0.4648
p=0.971
p=0.6316
p=0.2219
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

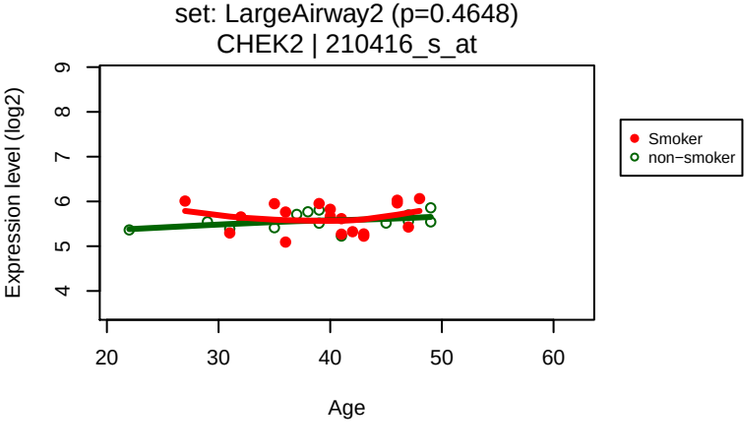

Gene expression ratio (smokers/nonsmokers)
gene: CHEK2 | 210416_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4679
p=0.6725
p=0.6687
p=0.5916
p=0.4648
p=0.971
p=0.6316
p=0.2219
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

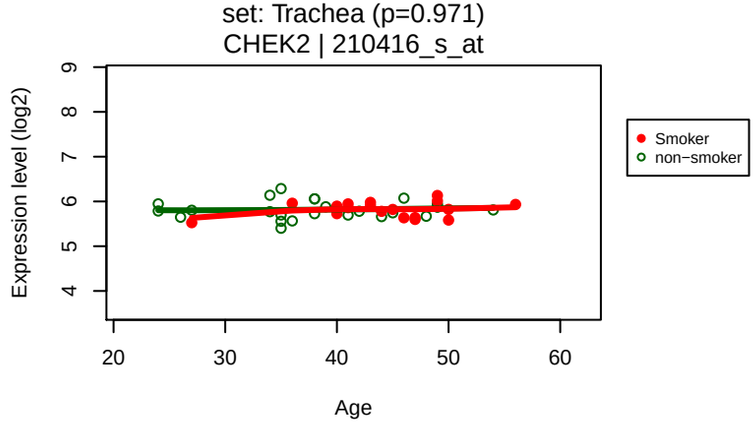

Gene expression ratio in Males
gene: CHEK2 | 210416_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9573
p=0.9061
p=0.5693
p=0.1662
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CHEK2 | 210416_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.617
p=0.1793
p=0.3118
p=0.9981
SmallAirway2
LargeAirway2
Trachea
Macrophage2

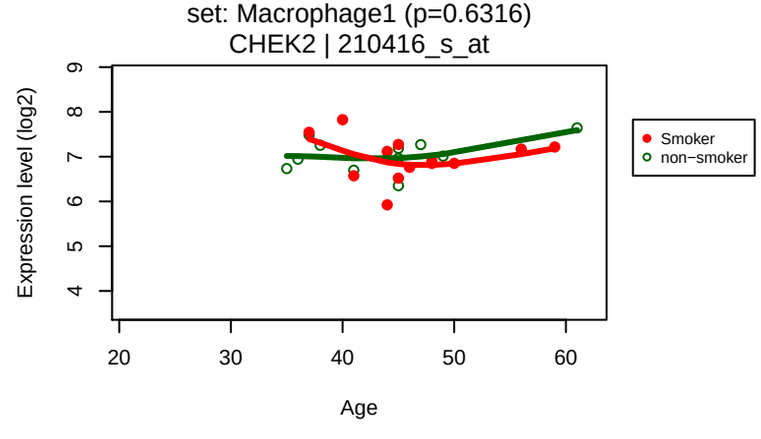

set: SmallAirway1 (p=0.4679)
CHEK2 | 210416_s_at
Expression level (log2)
Age
Smoker
non−smoker

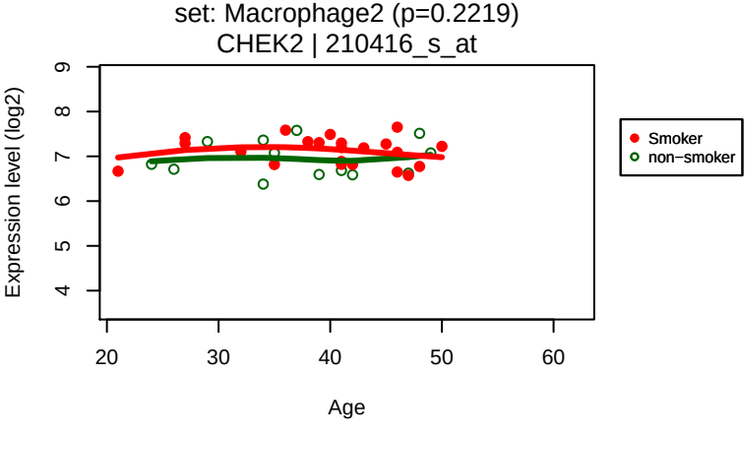

set: SmallAirway2 (p=0.6725)
CHEK2 | 210416_s_at
Expression level (log2)
Age
Smoker
non−smoker

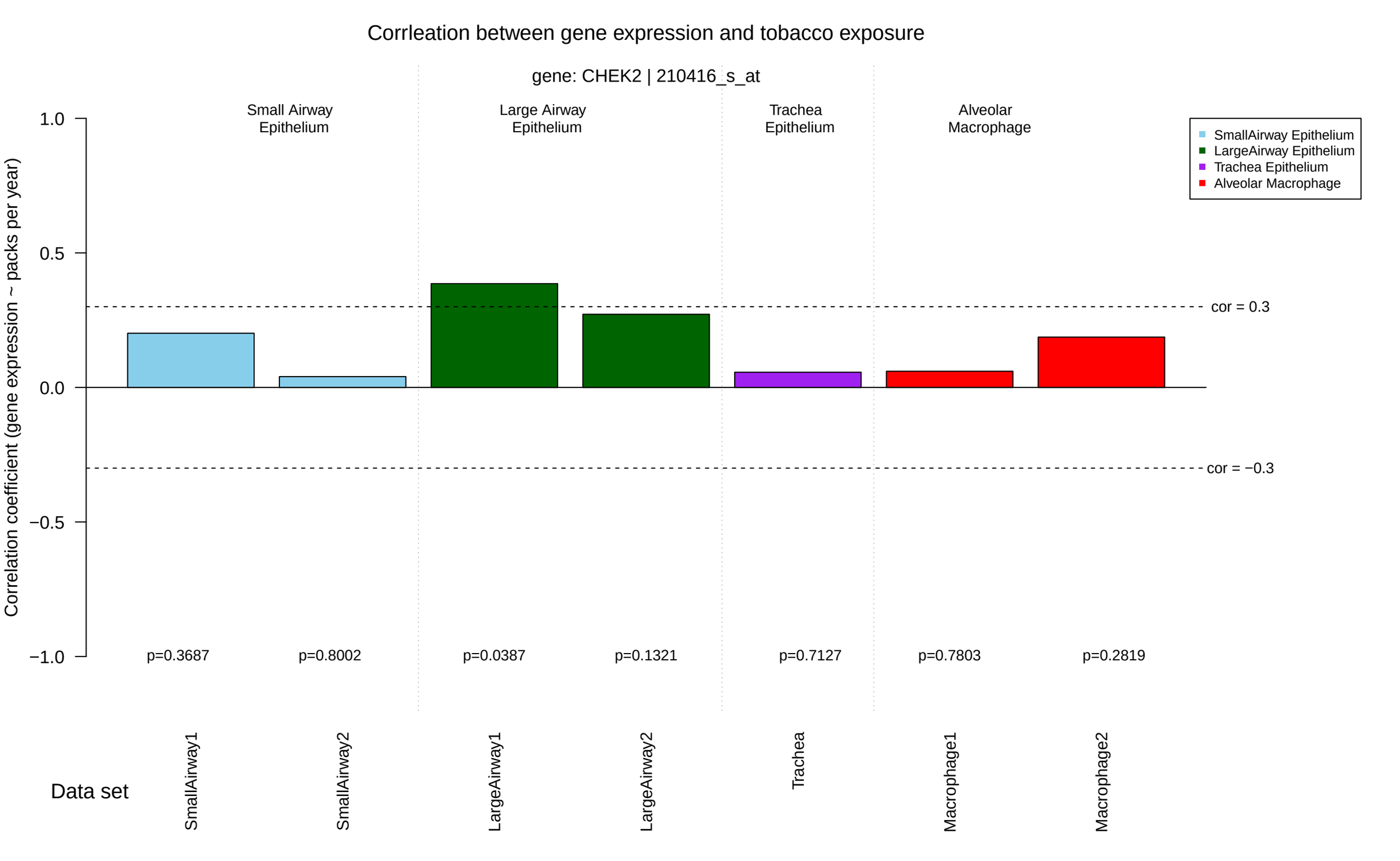

set: LargeAirway1 (p=0.5916)
CHEK2 | 210416_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4648)
CHEK2 | 210416_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.971)
CHEK2 | 210416_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6316)
CHEK2 | 210416_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2219)
CHEK2 | 210416_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CHEK2 | 210416_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3687
p=0.8002
p=0.0387
p=0.1321
p=0.7127
p=0.7803
p=0.2819
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2